БИОЛОГИЯ · 8 КЛАСС

УРОК 14. ВЛИЯНИЕ ЧЕЛОВЕКА НА ОКРУЖАЮЩУЮ СРЕДУ

Глава 14 учебника Biology Grade 8 (Astana-kitap, 2017). Оригинал: страницы PDF 351–386. После русской учебной части приложены исходные страницы с иллюстрациями.

Как работать: прочитай объяснение, изучи рисунки в приложении, выполни вопросы после подразделов и итоговую проверку. Проходной результат — 28 из 40.

Объяснение на русском

Деятельность человека меняет экосистемы и биоразнообразие. Для сохранения видов важны охраняемые территории, разумное использование ресурсов и снижение загрязнения. Экологические проблемы Казахстана включают вопросы состояния вод, почв и воздуха.

14.1 PRESERVING AND

Прочитай соответствующий подраздел в приложении. Найди основные определения, объясни роль изучаемых структур и сравни приведённые в учебнике примеры. Запиши важные английские термины вместе с русскими эквивалентами.

5 вопросов для самопроверки

• Какова основная идея этого подраздела?

• Какие новые биологические термины здесь введены?

• Какой пример из учебника иллюстрирует эту тему?

• Как связаны строение и функция описанных объектов?

• Чем отличаются два объекта или процесса, рассмотренные в разделе?

14.2 ECOLOGICAL PROBLEMS

Прочитай соответствующий подраздел в приложении. Найди основные определения, объясни роль изучаемых структур и сравни приведённые в учебнике примеры. Запиши важные английские термины вместе с русскими эквивалентами.

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ИТОГОВАЯ ПРОВЕРКА · 40 ВОПРОСОВ

Ответь на 40 вопросов по содержанию главы. Для заданий 1–20 используй оригинальные фрагменты учебника (английский язык); для 21–40 — объяснение и схемы. Оценивается содержательная правильность ответа.

1. Объясни по-русски смысл утверждения из учебника: “Stem cell: A type of cell that can turn into any other type of”

2. Объясни по-русски смысл утверждения из учебника: “Molecule: The smallest unit of a substance that is the”

3. Объясни по-русски смысл утверждения из учебника: “Urethra: the duct by which urine is conveyed out of the body”

4. Объясни по-русски смысл утверждения из учебника: “Microscope: Instrument used to observe things that are”

5. Объясни по-русски смысл утверждения из учебника: “or on objects that have been left for too long in warm, wet”

6. Объясни по-русски смысл утверждения из учебника: “Resource: A supply of something that can be used when”

7. Объясни по-русски смысл утверждения из учебника: “Chitin: a kind of polysaccharide which is found in the”

8. Объясни по-русски смысл утверждения из учебника: “1. What can you do to preserve the biodiversity as a student?”

9. Объясни по-русски смысл утверждения из учебника: “Capillaries are blood vessels that connect arteries and veins”

10. Объясни по-русски смысл утверждения из учебника: “has endured a long road to recovery, increasing population”

11. Объясни по-русски смысл утверждения из учебника: “Nucleic acid: In the nucleus of a cell, there are two major”

12. Объясни по-русски смысл утверждения из учебника: “A seed bank is a place where seeds of different plants are”

13. Объясни по-русски смысл утверждения из учебника: “Bacteria: The kingdom of life which has no cell membrane or”

14. Объясни по-русски смысл утверждения из учебника: “Veins: are blood vessels that transport blood to the heart”

15. Объясни по-русски смысл утверждения из учебника: “decreased. Below you can see water level change from 2005”

16. Объясни по-русски смысл утверждения из учебника: “Growth: The increase of size of an individual; can also be the”

17. Объясни по-русски смысл утверждения из учебника: “Lymphocyte: A lymphocyte is a type of white blood cell that”

18. Объясни по-русски смысл утверждения из учебника: “The world population is growing, and we must act fast as the”

19. Объясни по-русски смысл утверждения из учебника: “3. Which one of the following data are not related to”

20. Объясни по-русски смысл утверждения из учебника: “Mitochondria: An organelle in all eukaryotic cells which is”

21. Раздел 14.1 PRESERVING AND: Дай определение основному понятию темы.

22. Раздел 14.2 ECOLOGICAL PROBLEMS: Приведи пример из рисунка или текста.

23. Раздел 14.1 PRESERVING AND: Назови функцию одного из изученных объектов.

24. Раздел 14.2 ECOLOGICAL PROBLEMS: Сравни два понятия и укажи отличие.

25. Раздел 14.1 PRESERVING AND: Объясни, почему эта тема важна для живых организмов.

26. Раздел 14.2 ECOLOGICAL PROBLEMS: Дай определение основному понятию темы.

27. Раздел 14.1 PRESERVING AND: Приведи пример из рисунка или текста.

28. Раздел 14.2 ECOLOGICAL PROBLEMS: Назови функцию одного из изученных объектов.

29. Раздел 14.1 PRESERVING AND: Сравни два понятия и укажи отличие.

30. Раздел 14.2 ECOLOGICAL PROBLEMS: Объясни, почему эта тема важна для живых организмов.

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Проверка ответов

Вопросы открытые: сверь ответы с соответствующими утверждениями и иллюстрациями оригинального учебника. По 1 баллу за каждый содержательно верный ответ. 28/40 = 70%.

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CHAPTER 14.0

Human impact on environment



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14.1 PRESERVING AND MAINTAINING BIODIVERSITY

You will:

- discuss the reasons of preserving and maintaining biodiversity;

- understand the importance of seed banking.

Key terms

Conservation - looking after the environment so that animals and plants can live there.

Text

All living organisms have a certain place in the ecosystem. If human activities or natural disasters destroy their habitat, then they may not be able to survive. Example of some animals that humans hunted to extinction is Steller's Sea Cow, Great Auk.

Reasons to keep biodiversity of organisms:

1. Every species is unique, and each of them is important for ecology.

2. Preserving biodiversity supports important resources on a local and global scale.

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3. By conserving biodiversity, local politics can cooperate with nature savers to make income by using cultural resources.

We should make nature reserves and other protected areas, where people are not allowed to do any harm to the environment or to the animals and plants that live there.

To keep biodiversity of plants, many countries have developed seed banks.

A seed bank is a place where seeds of different plants are stored. There are about 1,400 seed banks around the world, but the most famous is the Svalbard Global Seed Vault, in Norway. It is home to nearly half a million specimens from around the world.

Seed banks are needed to store seeds.

Activity

A school nature reserve

Divide into three groups. Take 4 square meters of territory in your school. Then make a map of the territory. The best landscape plan will be a map according to which the class will plant plants. Then bring a plant from home and plant it in place according to the map. It can be a fruit tree or decoration flower and bushes. Then tell why you chose that plant to grow it in school. How can it be useful? Take care of your territory with your class.

Facts

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Przewalski's horse declined as its habitat was lost due to human interference. The last sighting of this wild horse was in the Gobi Desert of Mongolia in 1969.The Przewalski's horse has endured a long road to recovery, increasing population by only 15 to more than 400 in the wild today.

Research time

Tole bi is a Kazakh orator of senior zhuz who stayed alone when everyone moved during the invasion of jongars. Jongar 's kontayshi was angry with the fearless action of Tole bi and sent a man to him. Tole bi said that he could not move anywhere when he has a nest of swallows in his house. "It is not right to ruin somebody's home," he said. This was a great lesson to kontayshi who said "He is so wise. How can I ruin his home after all? Do not touch anybody of his auil" and turned back. Research proverbs and traditions about nature conservation.

Literacy

1. What can you do to preserve the biodiversity as a student?

2. Is it easy to preserve animal biodiversity or plant biodiversity? Explain your answer.

3. Do zoos help to preserve the biodiversity of animals?

Terminology

proverb - мақал-мәтел / пословица;

to destroy - жою / уничтожить;

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extinct - жойылып кеткен / вымерший;

disaster - апат / катастрофа;

biodiversity - көптүрлілік / биоразнообразие;

invasion - басып кіру / вторжение.

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14.2 ECOLOGICAL PROBLEMS OF KAZAKHSTAN

You will:

- Discuss the causes and solutions of environmentalProblems in Kazakhstan.

Key terms

Pollution - the condition in which air, soil, and water are contaminated by foreign substances.

STQ

What was the reason for closing Semey nuclear site by Nursultan Nazarbayev in 1989?

Text

Nature changed very much in the last century. The main reason for this is irresponsible human activities.

Growth in human population size leads to usage of more places for agriculture to supply people with food. They used many chemicals to kill insects protecting crop or minerals to make the crop grow. Also, human used agricultural lands for years without any opportunity for recovery of soil. This made soil polluted, and some lands even became desert.

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Also, they used rivers and other water resources for agriculture that decrease the volume of many seas and lakes where those rivers flow. For example, the water level of Aral sea decreased by 24 meters and reduced in the area when people used its rivers for agriculture. Some waters were polluted by wastes of factories like the Caspian Sea with oil.

Aral sea water levels in 1989 and 2014 Factories are polluting the air with emitted gases or heavy metals. For example, lead pollution in Shymkent, Oskemen, and Ekibastuz are results of industrial wastes. Heavy metals in air cause respiratory diseases and plant death. In water, heavy metals kill plants and animals living there.

There was nuclear testing in Semey which left radiation that makes water, soil and everything else dangerous to use due to mutation, etc.

All the pollution (air, water, soil, etc.) of nature lead to global natural disasters like climate change, hurricane, flood, and extinction of biodiversity.

Activity

All the garbage that we throw away can be used more than once. It is called recycling. Recycling is an important factor in the conservation of natural resources and the environment. The world population is growing, and we must act fast as the amount of waste we create is increasing all the time. It is necessary for us if we want to leave this planet beautiful for our future generations. So, let’s save our planet by recycling.

Divide into groups. Each group takes three big trash bags. One trash bag will be only for papers, second for plastics and last for cans or glasses.

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Design bags and put them in the school corners. Find recycling companies and give them filled bags for reuse.

Literacy

1. Describe a habitat near your house or school that is under threat from human activities and suggest what could be done.

2. Which city has the most ecologicalProblems in Kazakhstan? Defend your choice.

3. Which material can be recycled in your school?

Facts

To save Aral sea, Kazakhstan government built a dam Dike Kokaral in 2005. Dam is built between North Aral and South Aral, so Syrdariya water pours only into North Aral. Since then water level of North Aral has risen, and salinity is decreased. Below you can see water level change from 2005 to 2006.

Research time

Work in a group with four students.

Take any pollution or an ecological problem of Kazakhstan which

concerns you the most. Prepare a good exhibition stand which will increase awareness of this problem.

Organize a public presentation in school with your class. Here are some main ecologicalProblems of Kazakhstan:

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• radiation from nuclear testing sites • the reduction of the Aral Sea • desertification of former agricultural land

• pollution of the Caspian Sea with oil

• lead pollution of soil and water 151

Terminology

agriculture - ауыл шаруашылығы / селькое хозяйство;

emitted - шығарылған / выпущенный;

exhibition - көрме / выставка;

extinction - жоғалу, қырылып кету / вымирание;

garbage - қоқыс / мусор;

heavy metal - ауыр метал/тяжелый металл;

industrial - өнеркәсіптік / промышленный;

irresponsible - жауапкершіліксіз / безответственный;

nuclear - ядролық, атомдық / ядерный, атомный;

pollution - ластану, кірлену / загрязнение;

radiation - радиация;

recycling - қайта өңдеу / переработка.

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Problems

Test questions with one correct answer

1. Taking care about environment to make possible life for flora and fauna

A) Conservation

B) Ecology

C) Biodiversity

D) Pollution

E) Extinction

2. Show animal which is extinct because of human uncontrolled hunting

A) Steller's sea cows

B) Dinosaurs

C) Mammoth

D) Sable tube tigers

E) Dingo dogs

3. Lead pollution is high in

A) Kokshetau

B) Astana

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C) Zhezkazgan

D) Shymkent

E) Semey

4. Wild animal that was near to extinction because of human

A) Przewalski horse

B) Sea caws

C) Great Auk

D) Leopard seal

E) Wildebeest

Test questions with several (max 3) correct answers

1. Used for biodiversity conservation

A) Zoo

B) Botany parks

C) Oranjerie

D) Conservancy area

E) Seed bank

F) Red book

G) Black book

H) Dendropark

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2. Can affect negatively aquatic animals

A) Uncontrolled use of river water

B) Planting artificial forests

C) Building apartments

D) Acid rains

E) Smoke

F) Heavy rains

G) Using pesticides

H) Oil spills in oceans and seas

3. Which one of the following data are not related to Semipalatinsk Test Site?

A) 1949-1989 years

B) First Lighting

C) 29 August

D) 456 nuclear tests

E) Radiation

F) Hiroshima

G) 6 August 1945

H) Enola Gay

Matching

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1. Match terms with their definitions

    1. Seed bank

    2. Red book

    3. Conservancy area

A) Botany garden

B) Store seed to preserve genetic diversity

C) Preserve and enhance species

D) List of endangered species

E) Area for entertainment with animals

F) List of world’s popular species

2. Match pollutions with their causes

    1. Soil pollution

    2. Water pollution

    3. Air pollution

A) Planting trees

B) Using pesticides

C) Building zoos

D) Oil spills in oceans

E) Factories and cars

F) Recycling of garbage

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ANSWERS

1. CELL BIOLOGY	1	2	3	4
Answers for the test questions with one right answer	D	E	B	C
Answers for the test questions with several answers	B, F	B, E	A, B, F	
Matching questions	1-A, 2-D, 3-F	1-C, 2-B, 3-E		

2. CHEMISTRY OF LIFE	1	2	3	4
Answers for the test questions with one right answer	C	B	A	C
Answers for the test questions with several answers	B, D, G	B, D, E	A, B, C	
Matching questions	1-E, 2-D, 3-A	1-C, 2-D, 3-A		

3. DIVERSITY OF LIVING THINGS	1	2	3	4
Answers for the test questions with one right answer	A	B	A	E
Answers for the test questions with several answers	B, E, F	A, E, G	B, E, H	
Matching questions	1-B, 2-D, 3-E	1-B, 2-A, 3-F		

4. NUTRITION	1	2	3	4
Answers for the test questions with one right answer	C	E	E	A
Answers for the test questions with several answers	B, D, F	D, G, H	A, C, F	
Matching questions	1-C, 2-D, 3-F	1-E, 2-A, 3-C		

5. GROWTH AND DEVELOPMENT	1	2	3	4
Answers for the test questions with one right answer	C	D	B	A
Answers for the test questions with several answers	A, D, F	C, E, F	A, C, E	
Matching questions	1-B, 2-C, 3-E	1-D, 2-F, 3-B		

6. MATERIAL TRANSPORT	1	2	3	4
Answers for the test questions with one right answer	D	D	B	A
Answers for the test questions with several answers	B, D, F	C, E, G	A, D, H	
Matching questions	1-C, 2-D, 3-E	1-D, 2-B, 3-A		

7. RESPIRATION	1	2	3	4
Answers for the test questions with one right answer	E	C	E	D
Answers for the test questions with several answers	A, C, F	B, D, F	A, G, B	
Matching questions	1- E, 2-C, 3-D	1- B, 2-D, 3-F		

8. EXCRETION	1	2	3	4
Answers for the test questions with one right answer	B	A	D	B
Answers for the test questions with several answers	B, D, G	B, F, H	A, D, G	
Matching questions	1-C, 2-D, 3-E	1-B, 2-D, 3-E		

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9. MOVEMENT	1	2	3	4
Answers for the test questions with one right answer	B	A	C	C
Answers for the test questions with several answers	A, B, G	A, B, H	D, F, G	
Matching questions	1-A, 2-B, 3-D	1-A, 2-E, 3-F		

10. BIOPHYSICS	1	2	3	4
Answers for the test questions with one right answer	C	A	A	A
Answers for the test questions with several answers	B, D, F	B, D, E		
Matching questions	1-A, 2-B, 3-D	1-A, 2-E, 3-F		

11. COORDINATION AND REGULATION	1	2	3	4
Answers for the test questions with one right answer	E	B	B	B
Answers for the test questions with several answers	C, D, H	D, G, H	C, D, E	
Matching questions	1-E, 2-B, 3-C	1-B, 2-A, 3-D		

12. REPRODUCTION	1	2	3	4
Answers for the test questions with one right answer	D	E	D	A
Answers for the test questions with several answers	A, D, E	C, D,H	B, E, F	
Matching questions	1-C, 2-D, 3-E	1-A, 2-F, 3-C		

13. INHERITANCE AND VARIATION	1	2	3	4
Answers for the test questions with one right answer	C	D	A	D
Answers for the test questions with several answers	D, G	D, F	B, C, F	
Matching questions	1-E, 2-A, 3-C	1-A, 2-C, 3-E		

14. MICROBIAL DISEASES	1	2	3	4
Answers for the test questions with one right answer	D	A	B	E
Answers for the test questions with several answers	B, C, D	A, D, G	D, F, G	
Matching questions	1-A, 2-D, 3-E	1-F, 2-B, 3-A		

15. BIOSPHERE	1	2	3	4
Answers for the test questions with one right answer	D	A	C	D
Answers for the test questions with several answers	B, C, E	F, H, G	D, G, H	
Matching questions	1-D, 2-E, 3-A	1-A, 2-B, 3-F		

16. HUMAN IMPACT ON ENVIRONMENT	1	2	3	4
Answers for the test questions with one right answer	A	A	D	A
Answers for the test questions with several answers	D, E, F	A, B, H	B, E, H	
Matching questions	1-B, 2-D, 3-C	1-B, 2-D, 3-E		

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Glossary

A

Abdomen: The part of the body of a vertebrate containing the digestive and reproductive organs; the belly.

Absorption: Take up or take in; The sponge absorbs water well

Acid: A chemical that reacts easily with other substances and turns litmus paper red; less than 7 on the pH scale

Acne: inflammatory disease of the sebaceous gland

Active immunity: is immunity in an organism resulting from its own production of antibody or lymphocytes

Agglutination: adhesion of separate parts

Albinism: Albinism is a range of disorders varying in severity. They are all caused by a reduction or absence of the pigment melanin, often causing white skin, light hair, and vision problems

Alveoli: tiny air sacs found in a lung which is used for gas exchange

Amino Acid: A molecule that joins with other amino acids to form proteins

Analyze: Think about the different parts of a problem or situation to figure out how it is related to the whole.

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Angiosperms: flowering plants Arteriosclerosis: a disease of the arteries characterized by the deposition of fatty material on their inner walls.

Artery: Blood vessel that carries blood away from the heart Arthropods: animals with segmented body, jointed limbs and exoskeleton

Asexual Reproduction: Reproduction that happens without sex; one organism creates one or more organisms

Atom: The smallest unit of a substance that has all of the properties of that substance

Atrium: upper chamber of heart that receives blood from the body and lungs

B

Backbone: the row of connected bones that go down the middle of the back and protect the spinal cord.

B cells: a kind of lymphocyte that provide humoral immunity

Bacteria: The kingdom of life which has no cell membrane or nucleus and is always unicellular

Beriberi: a disease causing inflammation of the nerves and heart failure, ascribed to a deficiency of vitamin B1.

Bile: a yellow or greenish liquid that is made by the liver and that helps the body to digest fats.

Biological classification: Organization of how living things are related to each other

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Biology: The study of living things Blastula: early stage of development, two-layered ball like structure

Body cell: All of the cells in an organism not involved in reproduction

Bond: An electrical force that links atoms together

Bronchi: major air passages of the lungs which diverge from the windpipe.

C

Canines: a pointed tooth between the incisors and premolars of a mammal.

Calorie: Measure of energy, usually contained in food

Cancer: Disease where cells grow in an uncontrolled way

Capillaries are blood vessels that connect arteries and veins

Carbohydrate: An essential chemical in all cells that is broken down to form sugars; glucose, sucrose, lactose, galactose

Carbon Dioxide: Molecule made up of one carbon and two oxygens, produced by animals and other organisms; main contributor to manmade global warming

Cell: The basic structural and functional unit of all organisms

Cell differentiation: A process where cells change to perform different roles

Cell division: A process where one cell becomes more than one cell

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Cell membrane: Surrounds the cell and keeps it together; also decides what material enters and leaves

Cell organelle: Parts of the cell that perform specific functions

Cell wall: The stiff outer layer of a cell that protects the cell and gives it shape

Cellular respiration: The process where organisms get energy from organic molecules

Cellulose: a substance that is the main part of the cell walls of plants and that is used in making various products (such as paper)

Characteristic: A distinguishing quality of something; generosity is one of his best characteristics

Chloroplast: Organelle in plants and some other organisms which is responsible for photosynthesis

Chest cavity: internal space of chest where heart and lungs are located.

Chitin: a kind of polysaccharide which is found in the structure of fungi cell walls and insect skeleton.

Closed circulatory system: blood is contained inside blood vessels, circulating unidirectionally.

Compound: A chemical combination of two or more atoms (of different elements)

Consumer: Living thing that eats other living things

Cotyledon: part of the seed that will grow into the leafs.

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Covalent Bond: A type of chemical bond where electrons are shared between the atoms

Crop: organ used for storage of food in birds.

D

Decomposer: A type of living thing that survives by consuming dead organic matter

Denaturation: change of the structure of protein by heat or an acid.

Dentine: hard dense bony tissue forming the bulk of a tooth, under the enamel.

Development: growth from one cell to multicellular organism

Dermis: thick layer of skin under epidermis

Differentiation: Process where cells grow and become different than their mother cell

Diffusion: Where molecules spread out until they are evenly distributed in a medium, such as the air

Diaphragm: a large flat muscle that separates the lungs from the stomach area and that is used in breathing.

Diarrhea: an intestinal disorder that causes pass waste from your body very frequently and in liquid state

Dicot: plant with two cotyledons(seed leaves)

Digestion: breaking down food into small particles

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DNA: Abbreviation for deoxyribonucleic acid, which contains the genetic instructions for all forms of life

Dominant: A genetic characteristic that is always expressed by the organism

Donor: a person who gives some of their blood or a part of their body to help someone.

E

Ectoderm: the outermost layer of cells or tissue of an embryo in early development

Eczema: inflamed skin characterized by redness, itching, and scaly.

Edema: a condition characterized by an excess of watery fluid collecting in the cavities or tissues of the body.

Enamel: the white, compact, and very hard substance covering and protecting the dentin of a tooth.

Endoderm: the innermost layer of cells or tissue of an embryo in early development.

Embryology: The study of how organisms develop; the more closely related two organisms are, the more similar they are as they develop

Endoplasmic reticulum: Organelle that produces proteins and fats

Energy: The property of something's ability to do work

Environment: The complex of physical, chemical, and biotic factors (e.g., climate, soil, living things) that act upon an

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organism or an ecological community and ultimately determine their forms and survival

Epidermis: outermost layer of the skin Erythrocyte: are red blood cells, which carry oxygen

Esophagus: the part of the digestive system which connects the throat to the stomach.

Eukaryote: A type of organism that has a true nucleus in its cell(s)

Excretion: process of elimination of wastes from the body

Excretory system: a system that removes excess water and unnecessary products. It consists of kidneys, ureter, urinary bladder, and urethra.

Exhalation: movement of air out of lungs.

Exoskeleton: external skeleton that supports and protects an animal's body.

F

Fern: seedless, nonflowering vascular plant which lives in tropical to temperate regions.

Fiber: a thin, threadlike structure.

Fixed joint- joint between two bones that doesn't move

Fluid: Anything that flows; both gases and liquids are considered fluids

Food poisoning: becoming ill after eating spoiled food

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Forensics: science that use of scientific knowledge or methods in solving crimes

Fraternal twins: twins those are developed from two zygotes.

Fungi: The kingdom of living things that are eukaryotic and make their own energy; mushrooms, yeast, molds

G

Gametes: The cells that are responsible for sexual reproduction; sperm, eggs

Germ: A small organism that causes disease Gizzard: a muscular, thick-walled part of a bird's stomach for grinding food.

Gills: respiratory organ of aquatic animals.

Goggles: A safety device used whenever the eyes could be injured by a chemical or physical experiment

Golgi apparatus: Organelle that packages proteins and fats so that they can leave the cell

Growth: The increase of size of an individual; can also be the presence of something

Gymnosperm: a group of plant that have seeds unprotected by an ovary or fruit, including the conifers, cycads, and ginkgo.

H

Heart attack: blocking of heart arteries which brings to death of heart muscles

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Heterotroph: Living thing that needs to consume other organisms for food

Homeostasis: Maintenance of a constant internal environment in an organism.

Humoral immunity: antibody-mediated immunity

Hypertension: high blood pressure

Hypodermis: the deepest part of the skin which contains fat .ells

I

Identical twins: twins that are produced from a single zygote.

Incisor: tooth at the front of the mouth, adapted for cutting and biting.

Inflammation: physical condition in which part of the body becomes reddened, swollen, hot, and often painful.

Immunity: Ability to completely fight off disease

J

Joint: the area where two bones are attached for the purpose of permitting body parts to move

K

Kingdom: Most general classification of living things (sometimes placed under “Domains”)

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L

Large intestine: end part of the intestine that is wider and shorter than the small intestine.

Leukocyte: white blood cells, which protect the organism

Leukemia: Cancer of the blood or bone marrow

Lichen: A combination of a fungus and an algae that help each other

Life cycle: The series of stages in form and functional activity through which an organism passes between origin and expiration

Limbs: an arm or leg of a person or four-legged animal, or a bird's wing.

Lipid: Refers to a group of fats that cannot be dissolved in water

Liquid: The state of matter where the particles are loose and form the shape of their container but do not necessarily fill up the container

Living thing: An organism

Lymph: fluid that circulates through lymphatic system
Lymphatic system: network of tissues and organs that help rid the body of toxins, waste and other unwanted materials.

Lymphocyte: A lymphocyte is a type of white blood cell that is part of the immune system. There are two main types of lymphocytes: B cells and T cells.

Lysosome: Organelle that digests food and waste in the cell

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M

Macromolecules: a molecule containing a very large number of atoms, such as a protein, nucleic acid, or synthetic polymer.

Meiosis: The process of cell division which produces four sex cells (gametes) from one cell

Metamorphosis: A marked and more or less abrupt developmental change in the form or structure of an animal (e.g., butterfly or frog) occurring subsequent to birth or hatching

Microorganism: A small organism that is only visible underneath a microscope

Microscope: Instrument used to observe things that are smaller than can be seen with the naked eye

Micro preparations: prepared microscope slides

Mitochondria: An organelle in all eukaryotic cells which is responsible for energy production

Mitosis: The process of cell division which produces two body cells from one cell

Molar teeth: a grinding tooth at the back of a mammal's mouth.

Molecule: The smallest unit of a substance that is the combination of one or more atoms

Monera: Kingdom of life made up of bacteria (prokaryotes)

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Mold: a soft, green or grey growth that develops on old food or on objects that have been left for too long in warm, wet air.

Monocot: plant with one cotyledon

Monomer: identical units that join together to form polymer

Multicellular: Made up of more than one cell

Multicellular Organisms: Organisms that are made up of more than one cell

Muscle: a body tissue that can contract and produce movement

N

Nausea: sickness at the stomach, an involuntary impulse to vomit an atom; fission, fusion

Nonvascular plant: plants without vascular tissue such as mosses.

Nucleic acid: In the nucleus of a cell, there are two major types of nucleic acids: DNA and RNA Nucleotides: These are found on a strand of DNA or RNA as a sequence of bases

Nucleus: In biology, this refers to the middle of a cell; in physical science, this refers to the center of an atom

Nutrient: Any molecule that is needed for an organism to survive

O

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Objective: Based upon fact

Observation: Noticing or paying attention

Offspring: Children, of any organism

Open circulatory system: organisms with open circulatory system have open ended vessels. Blood leaves the vessels and enters to the body space.

Order: Classification of living things between Class and Family

Organ: Group of tissues that perform a certain function

Organ system: Group of organs that together perform a common function

Organelle: Part of a cell that performs a function for the cell

Organic: Comes from living things

Organic molecule: A molecule that contains carbon atoms bonded together

Organism: A living thing that can live and reproduce independently

Organogenesis: formation of organs

P

Passive immunity: the immunity resulting from the injection of antibodies or lymphocytes from another organism

Peristalsis: the involuntary constriction and relaxation of the muscles of the intestine or another canal, creating wave-like

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movements which push the contents of the canal forward.

Permanent teeth: second set of teeth in mammals that grows as the milk teeth are shed

Phagocytes: leukocytes which digest viruses and bacteria

Photosynthesis: The process that happens in plants and some other organisms which takes the sun's energy and turns it into usable energy; $6CO_2 + 6H_2O + Light = C_6H_{12}O_6$ (glucose) + $6O_2$

Plasma: liquid part of blood.

Pollen: the fertilizing element of flowering plants, consisting of fine, powdery, yellowish grains or spores, sometimes in masses.

Pollination: the transfer of pollen from the anther to the stigma.

Polymer: large molecule or macromolecule composed of many repeated parts

Premolar: situated in front of the molar teeth.

Producer: Living thing that makes its own energy from the sun

Prokaryote: A type of living thing that is single-celled and has no true nucleus

Protein: A sequence of amino acids

Protist: Kingdom of life made up of single-celled eukaryotes

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Pulmonary circulation: movement of blood from the heart to the lungs for oxygenation, then back to the heart again.

Pulp: soft part of the tooth where blood vessels and nerves are found.

Pulp cavity: the space within a tooth that contains the pulp.

Pulse: the regular beating of the heart, especially when it is felt at the wrist or side of the neck.

R

Reaction: When one or more substances are changed into other substances

Regeneration: The process that creates something over again

Reproduce: To create more of creating offspring

Reproduction: The process of creating offspring

Research: Discovering information that other scientists have already published

Resistance: (Biology) Ability to fight off some amount of disease

Respiratory minute volume: the amount of inhaled or exhaled air in a minute.

Resource: A supply of something that can be used when needed

Respiratory rate: rate of breathing

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Ribosome: Organelle that reads the mRNA to produce proteins

S

Sebaceous gland: (oil gland)a small gland in the skin which secretes a lubricating oily matter (sebum) into the hair follicles to lubricate the skin and hair.

Salivary glands: glands which produce saliva

Saturated: Completely full; for a solution, the most amount of solute that can be put into the solvent

Scabies: skin disease with itching and small raised red spots, caused by the itch mite.

Semi movable joints: joints between bones in which the motion is limited

Sexual reproduction: The combination of two individuals (genetically) to form one or more new organisms

Small intestine: narrow upper part of the intestine where digestion of food completes and absorption starts

Species: A very specific classification of organisms; all members of a species can mate together

Spleen: an abdominal organ involved in the production and removal of blood cells in most vertebrates and forming part of the immune system.

Starch: A kind of storage polysaccharide found in plants.

Stem cell: A type of cell that can turn into any other type of cell.

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Sternum: Flat bone that connects ribs to each other forming the front part of rib cage.

Stimulus: Anything that affects an organism

Stomach: saclike organ which store and digest food

Stroke: death of brain tissue occurred as a result of artery blockage

Substance: A type of matter that has the same properties; water, oxygen, carbon dioxide, diamond

Systemic circulation: movement of blood from the heart through the body to provide oxygen and nutrients to the tissues of the body while bringing deoxygenated blood back to the heart.

T

T cells: leukocytes that provide cell-mediated immunity

Tetter: any of various eruptive skin diseases

Tidal volume: volume of gas inhaled or exhaled in each respiration, during a normal, regular breathing

Tissue: Group of cells that perform a similar function

Tissue fluid: fluid between cells

Tonsil: either of two small masses of lymphoid tissue in the throat, one on each side of the root of the tongue.

Toxin: a poisonous substance and especially one that is produced by a living thing

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Trachea: a large membranous tube reinforced by rings of cartilage, extending from the larynx to the bronchial tubes and conveying air to and from the lungs; the windpipe.

Typhoid: an infectious bacterial fever with an eruption of red spots on the chest and abdomen and severe intestinal irritation.

U

Ultraviolet: A form of electromagnetic radiation that has more energy than visible light; most ultraviolet light is usually blocked in our atmosphere by ozone

Unicellular: Made up of one cell

Urinary bladder: a membranous sac in many vertebrates that serves for the temporary retention of urine and discharges by the urethra.

Urine: liquid waste excreted by the kidneys, in humans being a yellowish, slightly acid

Urea: a substance found in urine and also made from ammonia,

Ureter: a muscular duct or tube conveying the urine from a kidney to the bladder or cloaca

Urethra: the duct by which urine is conveyed out of the body from the bladder, and which in male vertebrates also conveys semen.

V

Vacuole: Organelle that stores nutrients in the cell

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Vascular: relating to or denoting the plant tissues (xylem and phloem) which conduct water, sap, and nutrients in flowering plants, ferns, and their relatives.

Vena cava: any of the large veins by which in air-breathing vertebrates the blood is returned to the right atrium of the heart.

Ventricle: one of two sections of the heart that pump blood out to the body.

Veins: are blood vessels that transport blood to the heart

Vessel: tube shaped structure that carries blood in the body.

Virus: A small particle that contains DNA or RNA and is able to reproduce only inside of a living cell

Vital capacity: greatest amount of air that can be forced from the lungs after maximum inhalation

Vitamin: organic molecule essential for body processes

W

Wax: an oily organic compound insoluble in water but soluble in organic solvents; essential structural component of living cells (along with proteins and carbohydrates)

Y

Yeast: an organism of the kingdom Fungi lacking chlorophyll and feeding on organic matter; ranging from unicellular or multicellular organisms to spore-bearing syncytia

Z

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Zygote: Combination of a sperm and egg cell

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